Signals and Systems

Arun B Aloshious

Laplace transform to Complex exponential signals

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- They have same period $T = \frac{2\pi}{\omega}$. No other bounded Eigen functions have this form.
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- They have same period $T = \frac{2\pi}{\omega}$. No other bounded Eigen functions have this form.
- This implies the signals of the form $\sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$ for all $a_k \in \mathbb{C}$ are periodic with period T.
- Can all periodic signals written of the form $\sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$ for some $a_k \in \mathbb{C}$?

Fourier Series- Continuous time

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

where

$$a_k = \frac{1}{T}\int_0^T x(t)e^{-j\frac{2\pi}{T}k}$$

Truncated version:

$$\hat{x}_N(t) = \sum_{k=-N}^{N} a_k e^{j\frac{2\pi}{T}k}$$

Convergence:

$$\hat{x}_N(t) \underset{M.S}{\overset{N\to\infty}{\longrightarrow}} x(t)$$

means $\frac{1}{T}\int_0^T |x(t) - \hat{x}_N(t)|^2 dt \to 0$

Fourier series of triangular wave

Examples

Some examples...

Example 1:

$$x(t) = \begin{cases} t & ; 0 < t \leq 1 \\ 2 - t & ; 1 < t \leq 2 \\ x(t+T) = x(t) & \text{for the rest} \end{cases}$$

Example 2:

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$$a_k = \begin{cases} 0.5 & ; k = 0 \\ \frac{(-1)^n - 1}{\pi^2 k^2} & ; k \neq 0 \end{cases}$$

Example 2:

$$x(t) = \begin{cases} t & ; 0 < t \leq 1 \\ x(t+T) = x(t) & \text{for the rest} \end{cases}$$

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Fourier series of square wave

Example

Some examples...

Example 3:

$$x(t) = \begin{cases} 1 & ; 0 < t \leq 1 \\ 0 & ; 1 < t \leq 2 \\ x(t+T) = x(t) & \text{for the rest} \end{cases}$$

Example 4:

$$x(t) = \begin{cases} 1 & ; 0 < t \leq 0.5 \\ 0 & ; 0.5 < t \leq 2 \\ x(t+T) = x(t) & \text{for the rest} \end{cases}$$

Fourier series of square wave

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Some examples...

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Example 4:

$$x(t) = \begin{cases} 1 & ;0 < t \leq 0.5 \\ 0 & ;0.5 < t \leq 2 \\ x(t+T) = x(t) & \text{for the rest} \end{cases}$$

Dirichlet's conditions

- Absolute integerability
- Finite maxima and finite minima
- Finite discontinuities.

Properties of Fourier series

- Linearity

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

where

$$a_k = \frac{1}{T}\int_{0}^{T} x(t) e^{-j\frac{2\pi}{T}k} dt$$

If $x_1(t) \overset{F.S.}{\leftrightarrow} a_k$ and $x_2(t) \overset{F.S.}{\leftrightarrow} b_k$, then
$\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{F.S.}$

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Properties of Fourier series

- Linearity
- Time shifting

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If $x(t) \overset{F.S.}{\leftrightarrow} a_k$, then

$x(t-t_0) \xleftrightarrow{F.S.} c_k = e^{-j\frac{2\pi}{T}kt_0}a_k$

Properties of Fourier series

- Linearity
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- Frequency shifting

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If $x(t) \overset{F.S.}{\leftrightarrow} a_k$, then
$e^{j\frac{2\pi}{T}tM}x(t) \xleftrightarrow{F.S.}$

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If $x(t) \overset{F.S.}{\leftrightarrow} a_k$, then

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Properties of Fourier series

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- Time shifting
- Frequency shifting
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- Conjugation

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Properties of Fourier series

- Linearity
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- Reflection
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- Periodic convolution

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

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If $x(t) \overset{F.S.}{\leftrightarrow} a_k$ and $h(t) \overset{F.S.}{\leftrightarrow} b_k$, then
$x(t) \circledast h(t) \xleftrightarrow{F.S.}$

Properties of Fourier series

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Properties of Fourier series

- Linearity
- Time shifting
- Frequency shifting
- Reflection
- Conjugation
- Periodic convolution
- Multiplication

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

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$$a_k = \frac{1}{T}\int_0^T x(t)e^{-j\frac{2\pi}{T}k}dt$$

If $x(t) \overset{F.S.}{\leftrightarrow} a_k$ and $y(t) \overset{F.S.}{\leftrightarrow} b_k$, then

$x(t).y(t) \xleftrightarrow{F.S.}$

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Properties of Fourier series

- Linearity
- Time shifting
- Frequency shifting
- Reflection
- Conjugation
- Periodic convolution
- Multiplication
- Parseval's theorem

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

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$\frac{1}{T}\int_0^T |x^2(t)|dt =$

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- Time shifting
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- Periodic convolution
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where

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If $x(t) \overset{F.S.}{\leftrightarrow} a_k$, then

$$\frac{1}{T}\int_0^T |x^2(t)|dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Properties of Fourier series

- Linearity
- Time shifting
- Frequency shifting
- Reflection
- Conjugation
- Periodic convolution
- Multiplication
- Parseval's theorem
- Some more

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}k}$$

where

$$a_k = \frac{1}{T}\int_0^T x(t)e^{-j\frac{2\pi}{T}k}dt$$

- x(t) is odd signal, a_k is odd
- x(t) is even signal, a_k is even
- x(t) is real signal, $a_{-k} = a_k^*$
- Some text denote a_k as $a[k]$.

Spectrum using Fourier series

- k versus a_k is called frequency spectrum of $x(t)$.

Example:

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- k versus $|a_k|$ is called magnitude spectrum of $x(t)$.

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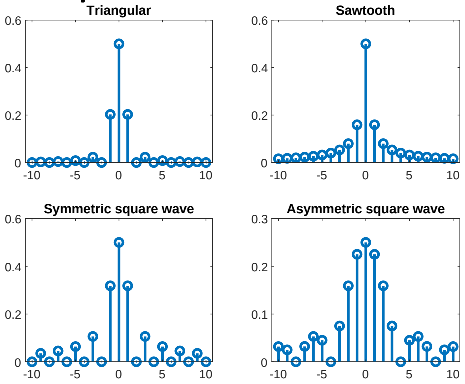

Spectrum using Fourier series

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- k versus $|a_k|$ is called magnitude spectrum of $x(t)$.
- k versus $angle(a_k)$ is called phase spectrum of $x(t)$.

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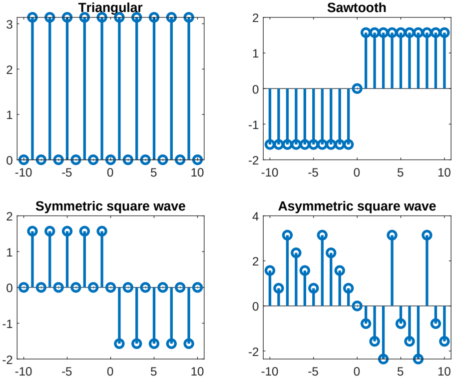
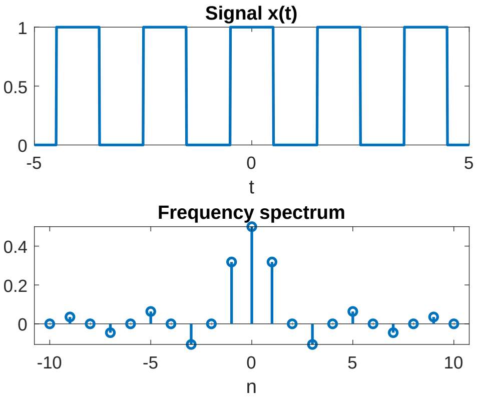


Fourier series- an example

1. $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$

Fourier series to Fourier transform

- Fourier series spectrum has additional parameter $\omega_0 = \frac{2\pi}{T}$.



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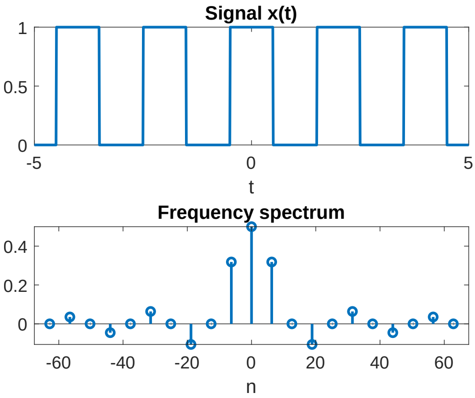

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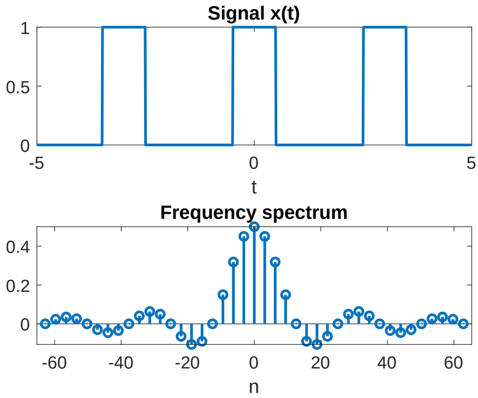
- What happens if T, the period is higher? (ω_0 small)



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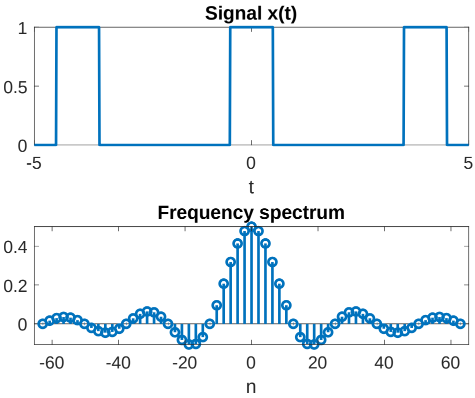
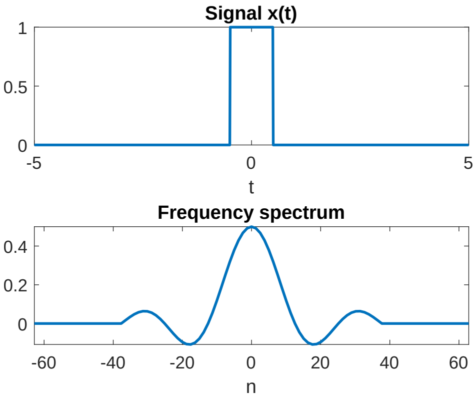


Fourier series to Fourier transform

- Fourier series spectrum has additional parameter $\omega_0 = \frac{2\pi}{T}$.
- What happens if T, the period is higher? (ω_0 small)
- $T \to \infty \Rightarrow$ Fourier transform
  $x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ where
  $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$



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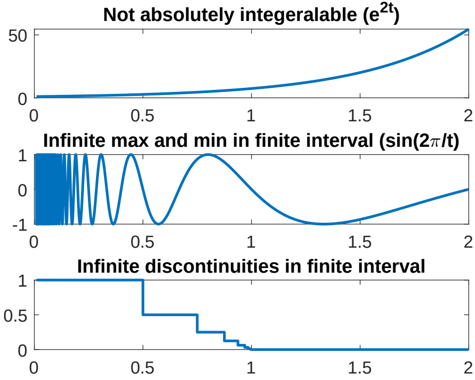
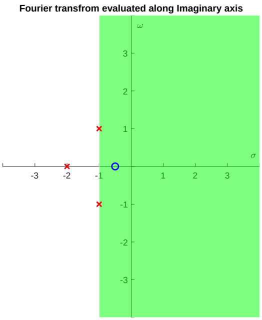
Existance $\Rightarrow$ Dirichlet's conditions

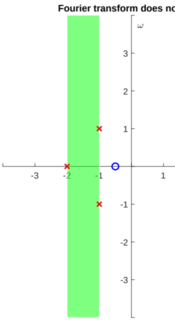
- Absolutely integerable $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
- $x(t)$ has finite discontinuities in finite interval
- $x(t)$ has finite maxima and finite minima in finite interval



Fourier transform and Laplace transform

1. $\mathcal{F}(x(t)) = \mathcal{L}(x(t))|_{s=j\omega}$ assuming Laplace transform exist. Otherwise Fourier transform does not exist because it violates Dirichlet's conditions.
2.





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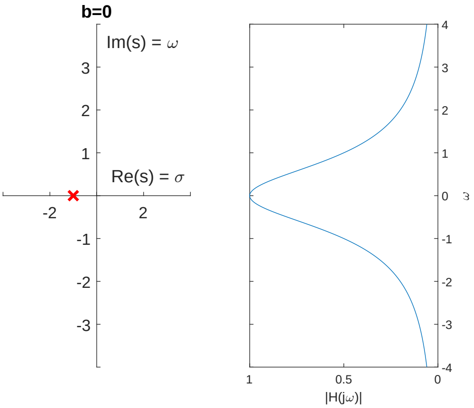
2. $H(s) = \frac{\alpha}{(s-0.5)^2}$



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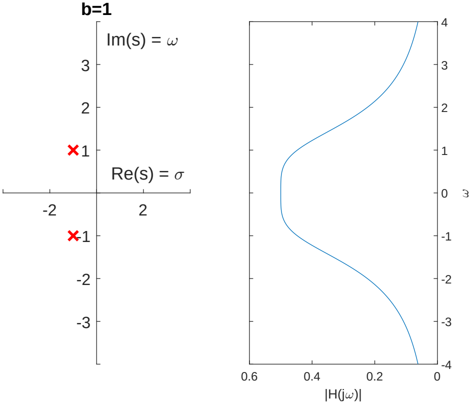

Fourier transform and Laplace transform

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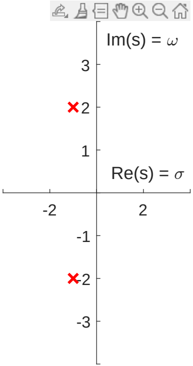
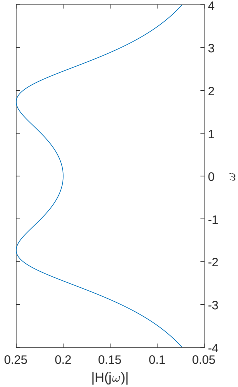
2. $H(s) = \frac{\alpha}{(s-0.5)^2+4}$





Fourier transform and Laplace transform

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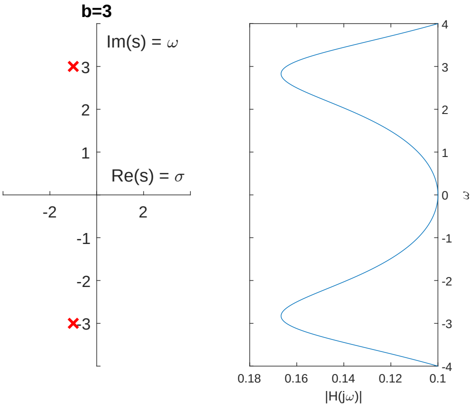
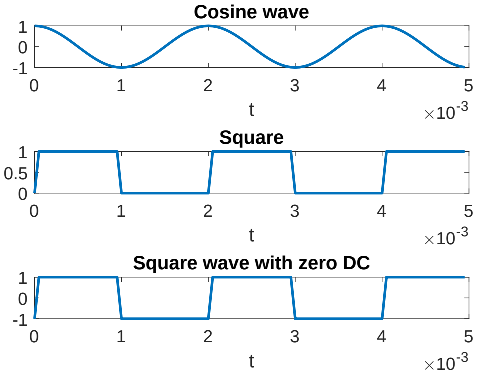
2. $H(s) = \frac{\alpha}{(s-0.5)^2+9}$



Fourier transform for periodic signals

1. Periodic signals are useful signals which are not absolutely integerable.



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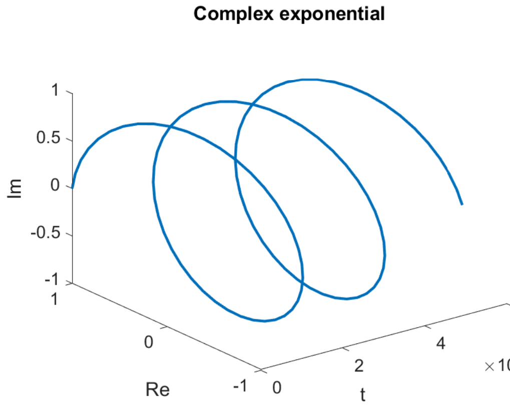

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2. Inverse Fourier transform of $2\pi\delta(\omega - \omega_0)$ is $e^{-j\omega_0 t}$. Thus,.

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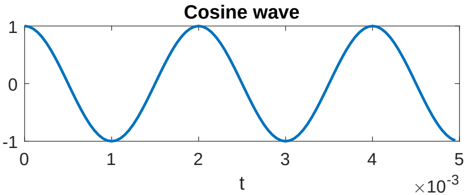
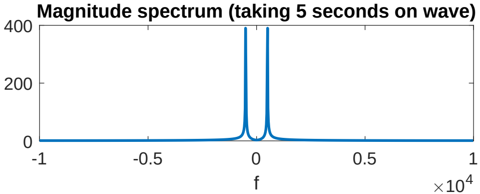
3. For periodic signals with Fourier series $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$, the Fourier transform is defined as

$$\mathcal{F}(x(t)) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - \frac{2\pi}{T}k\right)$$





Examples:

Rectangular pulse

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

Exponential

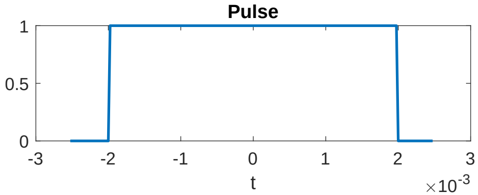
$$x(t) = e^{-t} u(t)$$

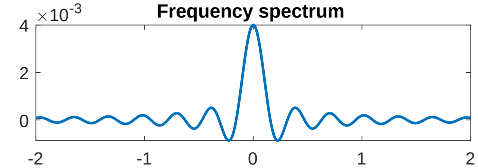
Examples:

Rectangular pulse

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$





Exponential

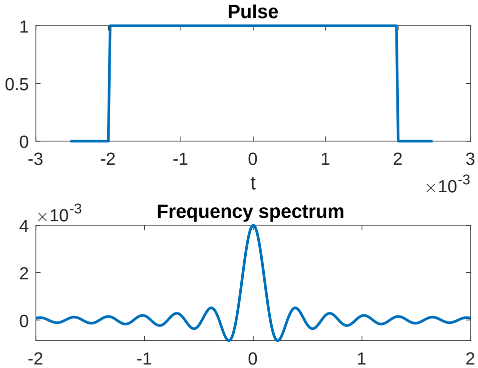
$$x(t) = e^{-t}u(t)$$

Examples:

Rectangular pulse

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

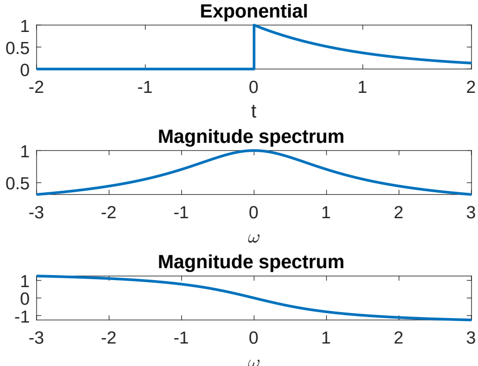
$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$



Exponential

$$x(t) = e^{-t}u(t)$$

$$X(\omega) = \frac{1}{1 + j\omega}$$



Examples:

Impulse signal

$$\delta(t)$$

Train of pulses

$$\sum_{k=-\infty}^{\infty} \delta(t - kT)$$

Properties of Fourier Transform - Linearity

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned} \mathcal{F}(x_1(t)) &:= X_1(j\omega) \\ \mathcal{F}(x_2(t)) &:= X_2(j\omega) \\ \alpha, \beta &\in \mathbb{C} \\ \Rightarrow \mathcal{F}(\alpha x_1(t) + \beta x_2(t)) &= \end{aligned}$$

Proof: Left as an exercise

Properties of Fourier Transform - Linearity

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned}
\mathcal{F}(x_1(t)) &:= X_1(j\omega) \\
\mathcal{F}(x_2(t)) &:= X_2(j\omega) \\
\alpha, \beta &\in \mathbb{C} \\
\Rightarrow \mathcal{F}(\alpha x_1(t) + \beta x_2(t)) &= \alpha X_1(j\omega) + \beta X_2(j\omega)
\end{aligned}$$

Proof: Left as an exercise

Properties of Fourier Transform - Time shifting

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned} \mathcal{F}(x(t)) &:= X(j\omega) \\ t_0 &\in \mathbb{R} \\ \Rightarrow \mathcal{F}(x(t-t_0)) &= \end{aligned}$$

Proof: Left as an exercise

Properties of Fourier Transform - Time shifting

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$

$$
\begin{aligned}
\mathcal{F}(x(t)) &:= X(j\omega) \\
t_0 &\in \mathbb{R} \\
\Rightarrow \mathcal{F}(x(t - t_0)) &= e^{j\omega t_0} X(j\omega)
\end{aligned}
$$

Proof: Left as an exercise

Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 1; & -3 \leq t \leq -1 \\ 1; & 1 \leq t \leq 3 \\ 0; & Elsewhere \end{cases}$$

Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

Find $\mathcal{F}(y(t))$ where

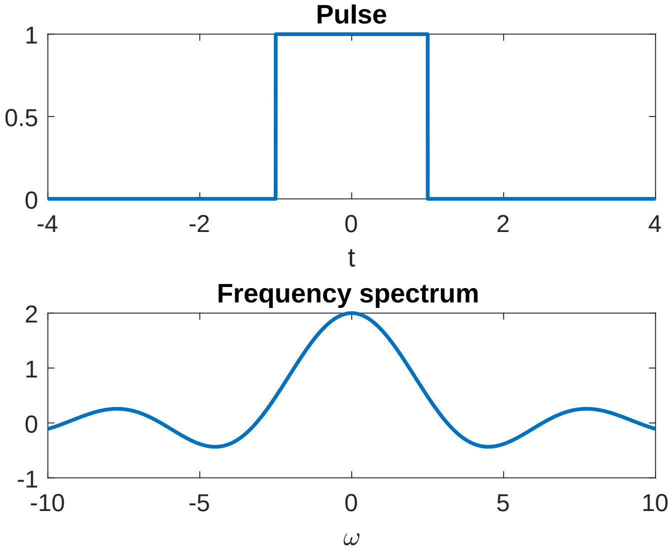
$$y(t) = \begin{cases} 1; & -3 \leq t \leq -1 \\ 1; & 1 \leq t \leq 3 \\ 0; & Elsewhere \end{cases}$$

Let $y_1(t) = y(t) = \begin{cases} 1; & -3 \leq t \leq -1 \\ 0; & Elsewhere \end{cases}$

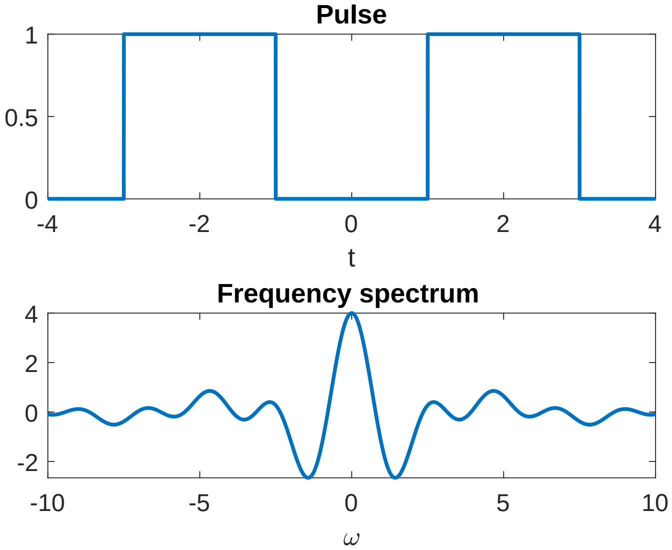
and $y_2(t) = y(t) = \begin{cases} 1; & 1 \leq t \leq 3 \\ 0; & Elsewhere \end{cases}$

$$\begin{aligned} y_1(t) &= x(t-2) \text{ with T} = 2 \text{ sec} \\ y_2(t) &= x(t+2) \text{ with T} = 2 \text{ sec} \\ y(t) &= y_1(t) + y_2(t) \\ Y(j\omega) &= Y_1(j\omega) + Y_2(j\omega) \\ &= e^{-2\omega} X(j\omega) + e^{2j\omega} X(j\omega) \\ &= 2cos(2\omega) X(j\omega) \\ &= \frac{4}{\omega} cos(2\omega) sin(\omega) \end{aligned}$$

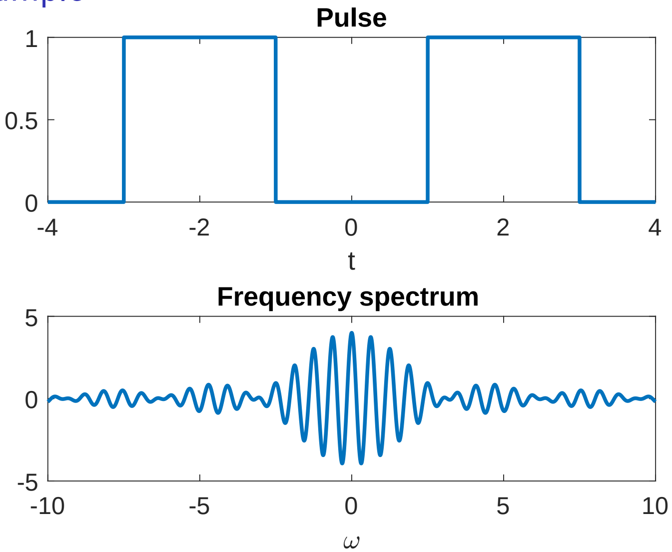
Example



Example



Example



Shifting by 10s

Properties of Fourier Transform - Frequency shifting

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned} \mathcal{F}(x(t)) &:= X(j\omega) \\ \omega_0 &\in \mathbb{R} \\ \Rightarrow \mathcal{F}(\qquad\qquad) &= X(j(\omega - \omega_0)) \end{aligned}$$

Proof: Left as an exercise

Properties of Fourier Transform - Frequency shifting

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$$\begin{aligned} \mathcal{F}(x(t)) &:= X(j\omega) \\ \omega_0 &\in \mathbb{R} \\ \Rightarrow \mathcal{F}(e^{-j\omega_0 t} x(t)) &= X(j(\omega - \omega_0)) \end{aligned}$$

Proof: Left as an exercise

Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 2cos(\omega_0 t); & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0; & Elsewhere \end{cases}$$

Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 2cos(\omega_0 t); & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0; & Elsewhere \end{cases}$$

$$\begin{aligned} y(t) &= e^{j\omega_0 t}x(t) + e^{-j\omega_0 t}x(t) \\ Y(j\omega) &= X(j(\omega + \omega_0)) + X(j(\omega - \omega_0)) \end{aligned}$$

Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

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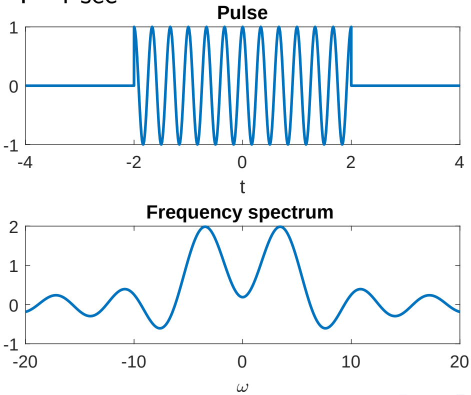
T=4 sec



Example:

Given

$$x(t) = \begin{cases} 1 & ; |t| \leq \frac{T}{2} \\ 0 & ; else \end{cases}$$

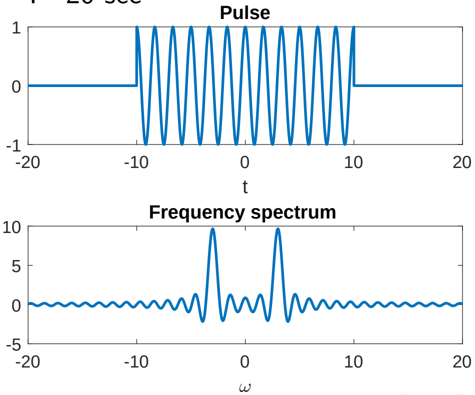
$$X(\omega) = \frac{2}{\omega} sin\left(\omega \frac{T}{2}\right)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 2cos(\omega_0 t); & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0; & Elsewhere \end{cases}$$

$$\begin{aligned} y(t) &= e^{j\omega_0 t}x(t) + e^{-j\omega_0 t}x(t) \\ Y(j\omega) &= X(j(\omega + \omega_0)) + X(j(\omega - \omega_0)) \end{aligned}$$

T=20 sec



Properties of Fourier Transform - Time scaling

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\begin{aligned} \mathcal{F}(x(t)) &:= X(j\omega) \\ a &\in \mathbb{R} \\ \Rightarrow \mathcal{F}(x(at)) &= \end{aligned}$$

Proof: Left as an exercise

Properties of Fourier Transform - Time scaling

$$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int\limits_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned} \mathcal{F}(x(t)) &:= X(j\omega) \\ a &\in \mathbb{R} \\ \Rightarrow \mathcal{F}(x(at)) &= \frac{1}{|a|} X\left(j\frac{\omega}{a}\right) \end{aligned}$$

Proof: Left as an exercise

Example

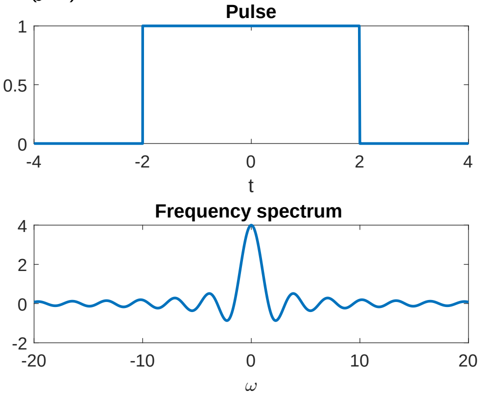
$$x(t) = \begin{cases} 1 & ; |t| \leq 2 \\ 0 & ; else \end{cases}$$

$$X(\omega) = \frac{2}{\omega} sin(\omega)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 1; & |t| \leq 4 \\ \\ 0; & Elsewhere \end{cases}$$

$X(j\omega)$



Example

$$x(t) = \begin{cases} 1 & ; |t| \leq 2 \\ 0 & ; else \end{cases}$$

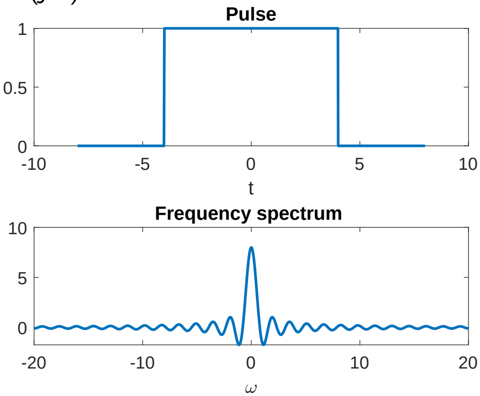
$$X(\omega) = \frac{2}{\omega} sin(\omega)$$

Find $\mathcal{F}(y(t))$ where

$$y(t) = \begin{cases} 1; & |t| \leq 4 \\ 0; & Elsewhere \end{cases}$$

$$y(t) = x(0.5t) \Rightarrow Y(j\omega) = \frac{1}{0.5} X(j\frac{\omega}{0.5})$$

$Y(j\omega)$



Properties of Fourier Transform - Convolution in time

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \\ Y(j\omega) &= X(j\omega)H(j\omega) \end{aligned}$$

Properties of Fourier Transform - Convolution in time

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \\ Y(j\omega) &= X(j\omega)H(j\omega) \end{aligned}$$

Proof:

$$\begin{aligned} Y(j\omega) &= \int_{t=-\infty}^{\infty} \int_{\tau=-\infty}^{\infty} x(\tau)h(t-\tau)d\tau e^{-j\omega t}dt \\ &= \int_{\tau=-\infty}^{\infty} x(\tau) \int_{t=-\infty}^{\infty} h(t-\tau)e^{-j\omega t}dtd\tau \\ &= H(j\omega) \int_{\tau=-\infty}^{\infty} x(\tau)e^{-j\omega\tau}d\tau \\ &= H(j\omega)X(j\omega) \end{aligned}$$

Properties of Fourier Transform - Time domain differentiation and integration

$$\mathcal{F}\left(\frac{dx(t)}{dt}\right) =$$

$$\mathcal{F}\left(\int_{-\infty}^{t} x(\tau)d\tau\right) =$$

Properties of Fourier Transform - Time domain differentiation and integration

$$\mathcal{F}\left(\frac{dx(t)}{dt}\right) = j\omega X(j\omega)$$

$$\mathcal{F}\left(\int\limits_{-\infty}^{t} x(\tau)d\tau\right) =$$

Properties of Fourier Transform - Time domain differentiation and integration

$$\mathcal{F}\left(\frac{dx(t)}{dt}\right) = j\omega X(j\omega)$$

$$\mathcal{F}\left(\int_{-\infty}^{t} x(\tau)d\tau\right) = \frac{1}{j\omega}X(j\omega)+\pi X(0)\delta(\omega)$$

Properties of Fourier Transform - Time domain differentiation and integration

$$\mathcal{F}\left(\frac{dx(t)}{dt}\right) = j\omega X(j\omega)$$

$$\mathcal{F}\left(\int_{-\infty}^{t} x(\tau)d\tau\right) = \frac{1}{j\omega}X(j\omega)+\pi X(0)\delta(\omega)$$

$$\begin{aligned}
\mathcal{F}(u(t)) &= \frac{1}{j\omega} + \pi\delta(\omega) \\
\int_{-\infty}^{t} x(\tau)d\tau &= \int_{-\infty}^{\infty} x(\tau)u(t-\tau)d\tau \\
&= x(t) * u(t) \\
\Rightarrow \mathcal{F}\left(\int_{-\infty}^{t} x(\tau)d\tau\right) &= \frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)
\end{aligned}$$

Properties of Fourier Transform - Parseval's relation

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$$\int\limits_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

Properties of Fourier Transform - Duality

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \text{ where } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\begin{aligned} \text{Let, } y(t) &= X(jt) \\ \mathcal{F}(y(t)) = \mathcal{F}(X(jt)) &= \end{aligned}$$

Properties of Fourier Transform - Duality

$x(t) = \frac{1}{2\pi} \int\limits_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ where $X(j\omega) = \int\limits_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$$\begin{aligned} \text{Let, } y(t) &= X(jt) \\ \mathcal{F}(y(t)) = \mathcal{F}(X(jt)) &= 2\pi x(-\omega) \end{aligned}$$

Properties of Fourier Transform - Multiplication in time

$$\mathcal{F}(x(t)y(t)) = \frac{1}{2\pi}\int\limits_{-\infty}^{\infty} X(j\Omega)Y(j(\omega-\Omega))d\Omega$$

$$\mathcal{F}(x(t)y(t)) = \frac{1}{2\pi}X(j\omega) * Y(j\omega)$$

Example

$$\sum_{n=-\infty}^{\infty} x(t)\delta(t - nT)$$

Example

$$\sum_{n=-\infty}^{\infty} x(t)\delta(t-nT)$$

$$\begin{aligned}
\mathcal{F}\left(\sum_{n=-\infty}^{\infty} \delta(t-nT)\right) &= \sum_{n=-\infty}^{\infty} e^{-j\omega nT} \\
&= \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi}{T}k) \\
\Rightarrow \mathcal{F}\left(\sum_{n=-\infty}^{\infty} x(t)\delta(t-nT)\right) &= \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega - \frac{2\pi}{T}k)
\end{aligned}$$

Properties-Summary

Property	Time Domain	Frequency Domain
Linearity	$a_1x_1(t)+a_2x_2(t)$	$a_1X_1(\omega)+a_2X_2(\omega)$
Time-Domain Shifting	$x(t-t_0)$	$e^{-j\omega t_0}X(\omega)$
Frequency-Domain Shifting	$e^{j\omega_0 t}x(t)$	$X(\omega-\omega_0)$
Time/Frequency-Domain Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{\omega}{a}\right)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Time-Domain Convolution	$x_1 * x_2(t)$	$X_1(\omega)X_2(\omega)$
Time-Domain Multiplication	$x_1(t)x_2(t)$	$\frac{1}{2\pi}X_1 * X_2(\omega)$
Time-Domain Differentiation	$\frac{d}{dt}x(t)$	$j\omega X(\omega)$
Frequency-Domain Differentiation	$tx(t)$	$j\frac{d}{d\omega}X(\omega)$
Time-Domain Integration	$\int_{-\infty}^{t}x(\tau)d\tau$	$\frac{1}{j\omega}X(\omega)+\pi X(0)\delta(\omega)$

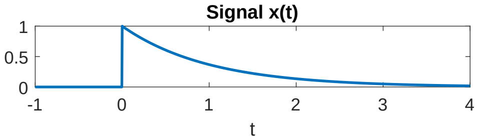
Property	
Parseval's Relation	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(\omega)\rvert^2 d\omega$
Even Symmetry	x is even $\Leftrightarrow$ X is even
Odd Symmetry	x is odd $\Leftrightarrow$ X is odd
Real / Conjugate Symmetry	x is real $\Leftrightarrow$ X is conjugate symmetric

Frequency, Magnitude and phase spectrum of Signals

- ω versus $X(j\omega)$ is called frequency spectrum of $x(t)$.

Example:

$$X(j\omega) = \frac{1}{1 + j\omega}$$



Frequency, Magnitude and phase spectrum of Signals

- ω versus $X(j\omega)$ is called frequency spectrum of $x(t)$.
- ω versus $|X(j\omega)|$ is called magnitude spectrum of $x(t)$.

Example:

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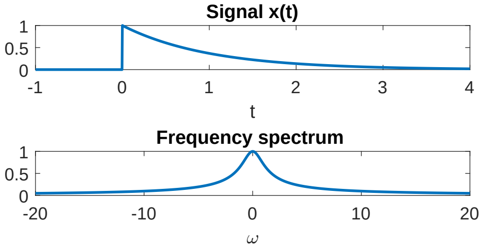
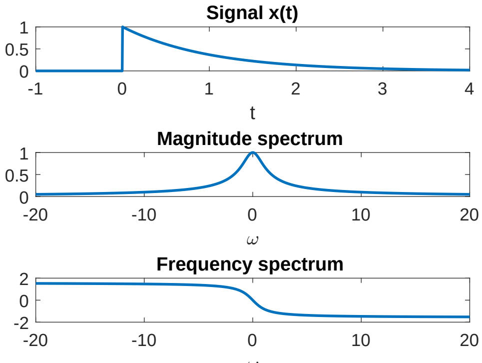


Frequency, Magnitude and phase spectrum of Signals

- ω versus $X(j\omega)$ is called frequency spectrum of $x(t)$.
- ω versus $|X(j\omega)|$ is called magnitude spectrum of $x(t)$.
- ω versus $angle(X(j\omega))$ is called phase spectrum of $x(t)$.

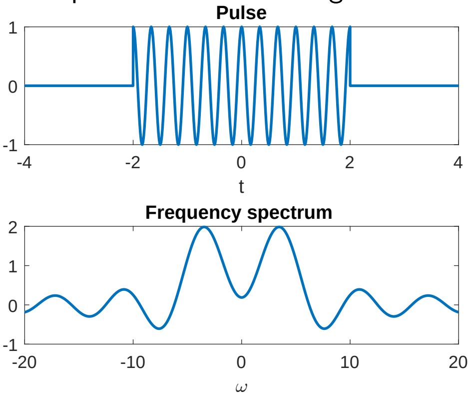
Example:

$$X(j\omega) = \frac{1}{1 + j\omega}$$



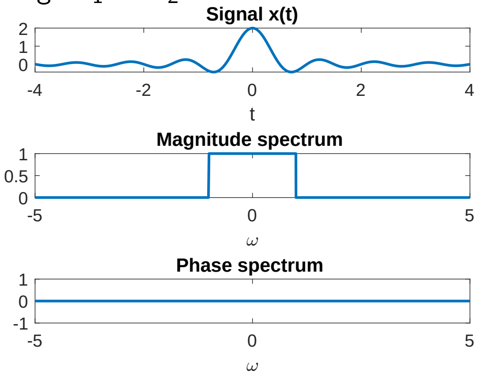
Band Limited signals

Example: A time limited signal and its frequency spectrum



Band Limited signals

A signal $x(t)$ is said to be band limited if the frequency spectrum is zero for all ω not in specific range ω_1 to ω_2.



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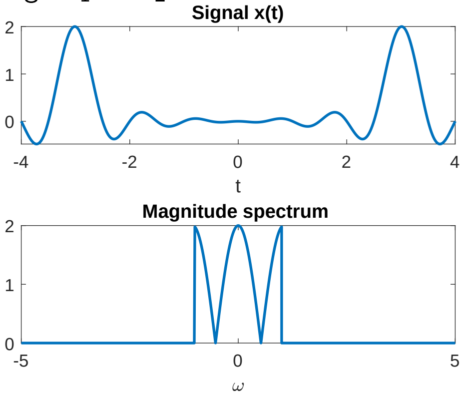

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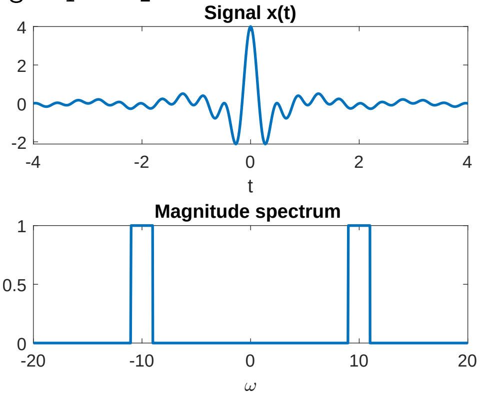

Example:

Frequency, Magnitude and phase response of LTI systems

Convolution Property: $y(t) = x(t) * h(t) \Rightarrow Y(j\omega) = H(j\omega)X(j\omega)$

Frequency, Magnitude and phase response of LTI systems

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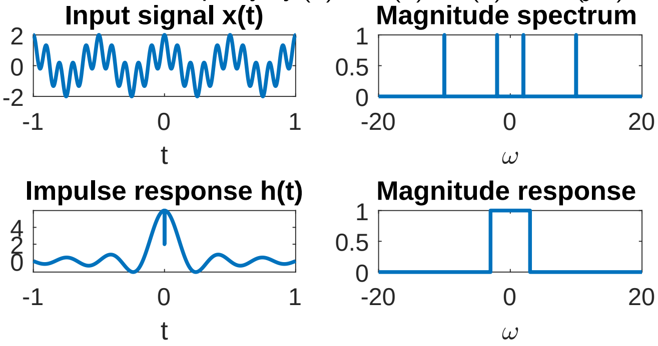
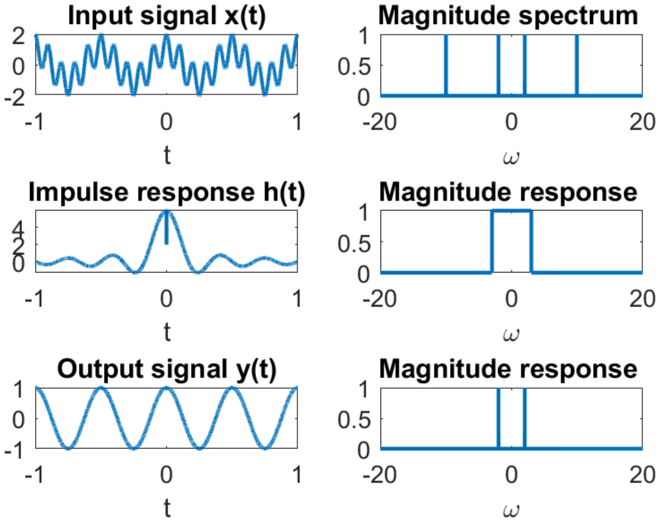


Frequency, Magnitude and phase response of LTI systems

Convolution Property: $y(t) = x(t) * h(t) \Rightarrow Y(j\omega) = H(j\omega)X(j\omega)$



Frequency, Magnitude and phase response of LTI systems

Convolution Property: $y(t) = x(t) * h(t) \Rightarrow Y(j\omega) = H(j\omega)X(j\omega)$
What happens to the phase of input? - Think

Gaussian wave

$$\begin{aligned} x(t) &= \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \\ \Rightarrow X(j\omega) &= \end{aligned}$$

Gaussian wave

$$
\begin{aligned}
x(t) &= \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \\
\Rightarrow X(j\omega) &= e^{-\omega^2/2}
\end{aligned}
$$

Fourier Transform - Time-Bandwidth relation

Less variance signal $\Rightarrow$ Large Variance in frequency domain
Less variance frequency domain $\Rightarrow$ Large Variance in time domain

$$
\begin{aligned}
\sigma_x^2 &= \frac{1}{\int\limits_{-\infty}^{\infty} |x(t)|^2 dt} \int\limits_{-\infty}^{\infty} (t - t_0)^2 |x(t)|^2 dt \\
\sigma_X^2 &= \frac{1}{\int\limits_{-\infty}^{\infty} |X(j\omega)|^2 d\omega} \int\limits_{-\infty}^{\infty} (\omega - \omega_0)^2 |X(j\omega)|^2 dt \\
\sigma_x^2 \sigma_X^2 &\geq \frac{1}{4}
\end{aligned}
$$